

Software Engineering Management and Economics

Experiment III Report

Economical Evaluation of Software Project

Course Project: HeartCompass (Digital Immortality System)

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Abstract

This report takes the course project HeartCompass (Digital Immortality System) as the evaluation object and, in accordance with the software project economical evaluation experiment requirements, analyzes its business goal, target users, market environment, cost structure, revenue model, taxes, cash flow, and economic evaluation indicators. HeartCompass is not positioned as a conventional chatbot or a static knowledge base. Instead, through long-term memory, user/persona profiles, vector retrieval, and large-model dialogue, it aims to gradually distill a "digital companion persona" capable of continuous, context-aware interaction.

Under the benchmark scenario, the five-year net present value (NPV) of the project is approximately RMB 438,700, the internal rate of return (IRR) is approximately 70.47%, and the dynamic investment payback period is approximately 2.33 years. Sensitivity analysis covering revenue decline of 20% combined with cost increase of 10% still yields a positive NPV. Therefore, based on the existing course project prototype, HeartCompass

demonstrates economic feasibility for continued refinement and small-scale product validation.

I. Experimental Purpose and Principles

1.1 Experimental Purpose

The purpose of this experiment is to design and complete a comprehensive economical evaluation report for a software project, using the course project as the evaluation object. Through this experiment, I need to: (1) master the modeling methods of software project evaluation; (2) understand software product pricing strategies; (3) record experimental procedures, organize experimental data, and establish parameter mapping relationships among financial tables; and (4) apply economic evaluation models to analyze the economic benefits and feasibility of software projects.

This report uses the HeartCompass course project as the evaluation object and completes the following work: clarifying the business goal of the project; collecting and organizing basic project information, market and economic environment data, and technical constraints; designing a business model and pricing strategy aligned with concrete user needs; estimating project size, investment, cost, and revenue; preparing auxiliary financial analysis statements; calculating project investment cash flow, NPV, IRR, and dynamic investment payback period; and ultimately judging the economic feasibility of the project.

1.2 Experimental Principles

The economical evaluation of a software project involves predicting the investment, revenue, cost, taxes, and cash flow of the project under the assumptions of the current fiscal and taxation system, market price system, and project life cycle, and then determining whether the project possesses profitability, investment recovery capability, and sustainable operation capacity.

This experiment adopts the project investment analysis approach (pre-financing analysis). It does not consider debt financing, interest expenses, or complex capital structures, but

evaluates profitability purely from the perspective of the cash flows generated by the project itself. The main evaluation indicators are:

- Net Present Value (NPV): $NPV = \sum (CI - CO)_t \times (1 + i_0)^{-t}$
- Internal Rate of Return (IRR): the discount rate that makes $NPV = 0$.
- Dynamic Investment Payback Period (P_t'): the time required for the cumulative discounted net cash flow to turn from negative to positive.

In the formulas above, CI represents cash inflow, CO represents cash outflow, i_0 represents the benchmark discount rate, and t represents the time period. If $NPV > 0$, $IRR > \text{benchmark discount rate}$, and $P_t' < \text{project life cycle}$, the project is generally considered economically feasible.

II. Course Project Overview and Basic Data

2.1 Basic Project Information

Project Element	Description
Project Name	HeartCompass (Digital Immortality / Digital Companion Persona System)
Project Type	AI-driven long-term memory and digital companionship service
Core Positioning	Distilling a continuously interactive digital companion persona through long-term memory, persona profiling, and context-aware dialogue
Main Form	CLI tool + backend service + Feishu (Lark) bot + vector database
Core Technologies	Python, LangChain/LangGraph, PostgreSQL/pgvector, Volcengine Ark LLM, Feishu Open Platform, Docker
Target Users	Individuals or small groups with needs for long-term companionship, memory preservation, and contextual continuity
Project Stage	MVP prototype completed; all 30 WBS tasks delivered; 298 person-hours

HeartCompass is the AI digital companion persona system developed in this course project. The core problem the project addresses is that conventional AI conversations typically focus only on single-session interactions and cannot stably remember a user's

long-term experiences, relationship changes, and important events. In practice, companionship needs require the system to continuously understand a person, remember past communications, form a relatively stable persona profile, and reflect continuity in subsequent conversations.

Around this goal, HeartCompass implements the architecture of "long-term memory + persona profile + dialogue entry point + periodic synchronization." Users complete configuration and interaction through the CLI and Feishu bot. The backend uses PostgreSQL/pgvector and LangGraph orchestration to preserve information and combine it with large-model inference to generate responses that are progressively closer to a specific person and relationship. The project has delivered a runnable engineering prototype encompassing CLI, backend services, database, model configuration, Feishu bot integration, logging, and test suites.

2.2 Business Goal

The business goal of HeartCompass is not to first build a generalized enterprise AI platform, but to validate whether the concrete demand for a "digital companion persona" is genuine. The project follows a three-stage roadmap:

- Stage 1 (MVP Validation): Complete the MVP version and seed-user testing, verifying installation, digital companion persona creation, long-term memory storage, persona profile synchronization, and Feishu bot interaction.
- Stage 2 (Subscription Launch): Launch subscription services around personal companionship and deep companionship, focusing on whether users are willing to pay for long-term memory, profile updates, and higher model invocation quotas.
- Stage 3 (Scenario Expansion): Provide shared memory spaces and customized commemorative archive services for families, small groups, graduation project teams, or research groups, thereby increasing average revenue per user in small-scale scenarios.

2.3 Market, Economic, and Social Environment Analysis

From the demand perspective, HeartCompass targets concrete needs for long-term companionship and memory accumulation rather than an abstract "AI tool market." Typical scenarios include: (1) users who wish to record personal growth and emotional

changes; (2) users who want to preserve stories, habits, and important milestones related to family and friends; (3) users who need AI to remember discussion contexts in long-term projects; and (4) users who need a companion entity capable of continuously understanding context in daily work and study.

From the economic perspective, the cost structure of a digital companion persona service primarily comes from servers, databases, model API calls, data storage, operations, and user support. Since the project leverages open-source components (LangGraph, PostgreSQL/pgvector) and cloud model services (Volcengine Ark), the initial hardware investment is moderate. However, as conversation frequency increases, model API costs will rise significantly. Therefore, the business model must incorporate invocation quotas, summary compression, caching mechanisms, and tiered model routing to control marginal costs.

From the social value perspective, the project can help users preserve long-term memories, reduce information loss, improve personal knowledge reuse efficiency, and provide lightweight memory archives for families and small teams. However, because the system handles chat records, persona profiles, and personal events, user authorization, data isolation, deletion mechanisms, encrypted storage, and privacy notices must be treated as mandatory constraints.

2.4 Technical and Economic Limitations

- **Technical limitation:** A digital companion persona requires long-term operation, continuous memory, and stable responses, so its deployment and maintenance complexity exceeds that of a one-time AI demo.
- **Cost limitation:** Model API costs are directly correlated with user conversation frequency; invocation quotas, summary compression, and caching are needed to control marginal cost.
- **Market limitation:** Willingness to pay for personal companionship products still requires validation. Early-stage small-scale seed-user testing should precede large-scale commercial rollout.
- **Compliance limitation:** Since personal memories, chat records, and persona profiles are

involved, the system must support user authorization, data deletion, permission isolation, and necessary privacy protection measures.

2.5 Data Collection Methodology

In accordance with the experiment requirement to "collect basic information about the market, economic & social environment, technical and economic limitations of the course project by online and offline," this report gathered data from the following sources:

- Online sources: (1) The HeartCompass GitHub repository (github.com/Jackey0903/HeartCompass) provided actual engineering data including 298 person-hours of WBS-tracked effort, 30 completed tasks, and NESMA function point measurements of 54 FP. (2) The 2024 China Software Industry Benchmark Data (CSBMK-202410) provided the regional labor rate of 32,129 RMB/Person-Month for Beijing. (3) Volcengine Ark platform documentation provided model API pricing data. (4) Public tax administration information provided current VAT (6%), corporate income tax (25%), and surcharge rates.
- Offline sources: (1) Course project documentation including README.md, BRIEF_INTRO.md, WBS_TRACKING.md, and five biweekly reports provided project scope, architecture, and progress data. (2) The "Economic Evaluation Methods and Parameters for Construction Projects (Third Edition)" (NDRC & MOC, 2006) provided the benchmark discount rate and evaluation methodology framework. (3) NESMA Function Point Analysis standards provided the size measurement framework.

All financial parameters used in this report are traceable to their sources. The blended labor rate of RMB 178.49/hour is derived from the CSBMK-202410 Beijing regional rate (32,129 RMB/Person-Month / 180 hours); the 10% benchmark discount rate follows the GB standard recommendation for software projects; and the tax rates reflect current PRC regulations. Project-specific data (workload, task count, function points) are extracted directly from the project's engineering records and cross-validated through dual-track analysis (NESMA vs. WBS backward tracking, variance = 5.20%).

III. Business Model and Pricing Strategy

This project adopts a tiered business model: "Free Trial → Personal Companion Subscription → Deep Companion Subscription → Family/Group Edition → Customized Commemorative Archive Service." This design is adopted because the value of a digital companion persona can only be demonstrated through sustained use. The free trial edition lowers the adoption barrier; the personal and deep companion editions generate stable subscription income; the family/group edition and customized archive service address stronger needs for memory preservation and sharing.

The pricing strategy combines value-based pricing with cost constraints. The Personal Companion Edition is priced accessibly to facilitate conversion. The Deep Companion Edition reflects higher value through expanded memory capacity and model invocation quota. The Family/Group Edition targets multi-user sharing scenarios with a higher average transaction value. The Customized Commemorative Archive Service is quoted based on data organization volume, deployment complexity, and service duration.

Product Version	Target Users	Core Rights	Pricing (Monthly)
Free Trial	First-time / demo users	1 persona, limited memory, basic dialogue	RMB 0
Personal Companion	Individual memory preservation	1-3 personas, long-term memory, standard quota	RMB 19.9
Deep Companion	Heavy users / knowledge workers	More personas, higher quota, priority responses	RMB 49.9
Family/Group Edition	Families, project teams	Shared memory space, multi-user permissions	From RMB 199
Customized Archive	Memorial / graduation needs	Data organization, private deployment, custom output	From RMB 3,000/project

IV. Economic Evaluation Model and Key Assumptions

For the purpose of course experiment calculation, this report adopts a five-year project life cycle. The construction period is treated as Year 0, and the operation period spans

Year 1 through Year 5. All monetary amounts are expressed in RMB 10,000 (ten thousand yuan) unless otherwise noted. This report does not incorporate debt financing or equity financing plans, but analyzes the investment, operating revenue, cost, and cash flow of the project itself (pre-financing analysis).

Note: The actual cash expenditure of a course project may be lower than that of a fully commercialized project. However, economic evaluation must account for opportunity cost. Therefore, this report converts student development time, testing documentation, deployment configuration, and project management effort into commercialized cost equivalents, so as to produce an evaluation result closer to real-world software project decision-making.

Parameter	Value	Description
Calculation Period	5 years	Operation cycle after MVP delivery
Benchmark Discount Rate (i_0)	10%	Based on software industry average cost of capital
VAT Rate	6%	Applicable to software services (general taxpayer)
Surcharges	12% of VAT	Urban maintenance & construction tax + education surcharge + local education surcharge
Corporate Income Tax Rate	25%	Standard PRC corporate income tax rate
Fixed Asset Depreciation	5-year straight-line, 0 residual	Servers, testing equipment, deployment infrastructure
Intangible Asset Amortization	5-year straight-line, 0 residual	Capitalized software development costs (code, documentation, configurations)
Residual Value	0	Conservative estimation; no separate salvage value assumed

V. Project Size and Investment Estimation

5.1 Project Size Estimation

According to the actual engineering records of the course project, HeartCompass has completed all 30 WBS tasks spanning requirements analysis, core function development, database and model configuration, Feishu bot integration, user documentation, and testing and validation. The project records an accumulated workload of 298 person-hours. Applying a commercialized blended labor rate of approximately RMB 120 per person-hour (aligned with junior-to-mid-level development and testing comprehensive cost in the Beijing software industry), the direct development labor cost is approximately RMB 35,800.

Additionally, the project's functional size was measured using the NESMA function point analysis method at 54 FP (comprising 7 ILF + 10 ILF + 7 ILF + 5 ELF + 4 EI + 6 EI + 7 EO + 5 EO + 3 EQ). A WBS-driven backward tracking cross-check confirmed the cost estimate with a variance of only 5.20%, validating high internal consistency. Considering supplementary costs for requirements analysis, system design, testing, documentation, deployment, security review, and project management, this report estimates the total capitalized software development investment at RMB 65,000.

Size / Workload Item	Estimated Value	Description
WBS Tasks Completed	30 tasks	Full decomposition from requirements to deployment
Accumulated Workload	298 person-hours	Requirements, development, debugging, documentation, testing, PM
Blended Labor Unit Cost	RMB 120 / hour	Commercialized opportunity cost for junior-to-mid-level engineers
Direct Development Labor Cost	RMB 35,800	298 hours × RMB 120 / hour
Capitalized Software Development Investment	RMB 65,000	Includes supplementary costs for requirements, design, testing, deployment, security, and project management

NESMA Functional Size	54 FP	Measured via NESMA FPA; cross-checked against WBS with 5.20% variance
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5.2 Investment Estimation Table

Investment Item	Amount (RMB 10,000)	Estimation Basis
Software Development (Intangible Asset)	6.50	Core code, persona profiles, conversation engine, deployment scripts
Server / Testing Equipment / Deployment	1.50	Testing server, database, vector retrieval, container infrastructure
Testing, Compliance, and Launch Preparation	1.00	Security audit, privacy notice, user manual, trial launch
Initial Working Capital	3.00	Cloud resources, model API calls, and short-term operating turnover
Total Project Investment	12.00	One-time investment during the construction period (Year 0)

5.3 Working Capital Estimation Table

Item	Year 0	Year 1	Year 2	Year 3	Year 4	Year 5
Working Capital Investment	3.00	0	0	0	0	0
Working Capital Recovery	0	0	0	0	0	3.00

Working capital is primarily used for cloud resource prepayment, model API invocation fees, database storage costs, and short-term turnover during small-scale promotion. This report assumes RMB 30,000 of working capital is injected in Year 0 and fully recovered at the end of Year 5.

5.4 Fixed Asset Depreciation and Intangible Asset Amortization Table

Item	Original Value (RMB 10,000)	Period	Year 1	Year 2	Year 3	Year 4	Year 5
Fixed Asset Depreciation	1.50	5 years	0.30	0.30	0.30	0.30	0.30
Intangible Asset Amortization	6.50	5 years	1.30	1.30	1.30	1.30	1.30
Total	8.00	—	1.60	1.60	1.60	1.60	1.60

5.5 Fund Raising Plan

The total initial investment of RMB 120,000 for HeartCompass is relatively modest by software startup standards. This report proposes a self-funded + angel investment model consistent with the project's nature as an MVP-stage AI companionship service:

- Self-Funding (RMB 40,000, 33.3%): The founding team contributes development equipment, existing code assets (valued at cost), and initial operating expenses. Since the course project prototype has already been developed, the capitalized software development cost (RMB 65,000) represents an in-kind contribution from the team's prior effort.
- Angel / Seed Round (RMB 80,000, 66.7%): To cover server procurement (RMB 15,000), testing and compliance preparation (RMB 10,000), and the first year of working capital (RMB 30,000), with the remaining RMB 25,000 as contingency reserve. Given the project's focused positioning, demonstrable MVP, and modest capital requirement, this amount is within the typical range for pre-seed / angel rounds in the Chinese AI startup ecosystem.

The fund raising plan is structured to minimize dilution: the team's existing development work (298 person-hours, equivalent to RMB 35,800 in direct labor cost) serves as the primary in-kind contribution. External funding is sought only for cash-intensive items (hardware, paid cloud services, and initial working capital). This structure preserves

founder equity while ensuring adequate cash runway for the first 12-18 months of operation.

Should external funding not materialize, the project can adopt a bootstrapping path: defer server purchases by using existing development machines, reduce working capital to RMB 15,000 by limiting initial model API consumption to free-tier quotas, and launch with only the Personal Companion Edition to generate early revenue. Under this bootstrap scenario, the total initial cash outlay drops to approximately RMB 25,000, which is independently financeable by the founding team.

5.6 Capital Usage Plan

The capital usage plan details the allocation and timing of the RMB 120,000 total investment across the construction period (Year 0) and early operation period. The plan ensures that funds are deployed in alignment with project milestones:

Capital Usage Item	Amount (RMB 10,000)	Timing	Milestone Trigger
Software Asset Capitalization	6.50	Year 0, Month 1	MVP completion and code freeze
Server & Infrastructure Procurement	1.50	Year 0, Month 1-2	Deployment environment provisioning
Testing & Compliance Preparation	1.00	Year 0, Month 2-3	Security audit and launch readiness
Working Capital — Cloud & API Prepayment	1.50	Year 0, Month 3	Service launch and first user onboarding
Working Capital — Operating Reserve	1.00	Year 0, Month 3-6	Buffer for initial operations
Working Capital — Promotion & Content	0.50	Year 0, Month 4-6	Seed-user acquisition and content organization
Total	12.00	Year 0 (6 months)	—

The capital deployment follows a phased approach: Months 1-2 focus on asset formation (software capitalization, server procurement); Months 2-3 cover launch preparation (testing, compliance, documentation); Months 3-6 fund initial operations (cloud services, model API calls, seed-user promotion). The RMB 30,000 working capital component is distributed across cloud prepayment (RMB 15,000), operating reserve (RMB 10,000), and promotion (RMB 5,000) to ensure that the service can operate sustainably from launch through the first revenue-generating period without additional cash injections.

VI. Revenue, Taxes, and Cost Estimation

6.1 Operating Revenue Estimation

The revenue projection is grounded in the actual usage scenarios of digital companion personas. In Year 1, revenue primarily originates from seed users, small-scale trial conversions, and a limited number of customized commemorative archive projects. From Year 2 through Year 5, as product stability improves, memory effects strengthen, and family/group scenarios expand, subscription revenue from personal plans, family/group editions, and customized services grows year over year.

Revenue assumptions are conservative: the free trial to paid conversion rate is estimated at 3-5%, average revenue per paying user (ARPU) at approximately RMB 30/month, and user growth follows a gradual organic trajectory rather than an aggressive exponential curve.

Item	Year 1	Year 2	Year 3	Year 4	Year 5
Personal / Deep Companion Subscription	5.50	12.00	22.00	34.00	48.00
Family / Group Edition	2.50	5.00	9.00	14.00	20.00
Customized Commemorative Archive Service	2.00	4.00	7.00	10.00	12.00
Total Operating Revenue	10.00	21.00	38.00	58.00	80.00

6.2 VAT and Surcharge Estimation Table

Item	Year 1	Year 2	Year 3	Year 4	Year 5
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Operating Revenue (tax-inclusive basis)	10.00	21.00	38.00	58.00	80.00
VAT Payable (6%)	0.60	1.26	2.28	3.48	4.80
Urban Maintenance & Construction Tax (7% of VAT)	0.04	0.09	0.16	0.24	0.34
Education Surcharge (3% of VAT)	0.02	0.04	0.07	0.10	0.14
Local Education Surcharge (2% of VAT)	0.01	0.03	0.05	0.07	0.10
Total Surcharges (12% of VAT)	0.07	0.15	0.27	0.42	0.58

6.3 Total Cost and Expense Estimation Table

Cost Item	Year 1	Year 2	Year 3	Year 4	Year 5
Cloud Resources and Model API Cost	2.50	4.20	7.00	10.00	14.00
Operations, Maintenance, and User Support	2.00	3.50	5.50	8.00	10.00
Seed-User Promotion and Content Organization	1.50	2.50	4.50	6.50	8.00
Management, Compliance, and Office Expenses	0.80	1.50	2.50	4.00	5.50
Subtotal — Operating Costs	6.80	11.70	19.50	28.50	37.50
Fixed Asset Depreciation	0.30	0.30	0.30	0.30	0.30
Intangible Asset Amortization	1.30	1.30	1.30	1.30	1.30
Total Costs and Expenses	8.40	13.30	21.10	30.10	39.10

Cost structure analysis: The largest cost component is cloud resources and model API fees, accounting for approximately 35-37% of total costs, which is characteristic of AI-native software services. Operations and maintenance constitute the second-largest category. Depreciation and amortization together represent a declining share of total costs as revenue scales, reflecting the improving operating leverage of the business model.

VII. Project Investment Cash Flow Analysis

This section evaluates the profitability of the project itself based on project investment and operating cash flows (pre-financing analysis). In the cash flow statement, Year 0 cash outflow is the total initial investment of RMB 120,000. Operating cash flow for Years 1-5 is calculated as Net Profit + Depreciation & Amortization Add-back + Working Capital Recovery (Year 5 only).

7.1 Project Investment Cash Flow Statement

Item	Year 0	Year 1	Year 2	Year 3	Year 4	Year 5
1. Cash Inflow	0	10.00	21.00	38.00	58.00	80.00
Operating Revenue	0	10.00	21.00	38.00	58.00	80.00
Working Capital Recovery	0	0	0	0	0	3.00
2. Cash Outflow	12.00	7.25	13.74	23.93	35.79	45.16
Initial Investment	12.00	0	0	0	0	0
Operating Costs	0	6.80	11.70	19.50	28.50	37.50
Surcharges	0	0.07	0.15	0.27	0.42	0.58
Income Tax	0	0.38	1.89	4.16	6.87	10.08
3. Net Cash Flow (1-2)	-12.00	2.75	7.26	14.07	22.21	34.84
4. Discount Factor (i=10%)	1.0000	0.9091	0.8264	0.7513	0.6830	0.6209
5. Discounted Net Cash Flow	-12.00	2.50	6.00	10.57	15.17	21.63
6. Cumulative Discounted NCF	-12.00	-9.50	-3.50	7.07	22.24	43.88

Income tax calculation detail: Total Profit = Operating Revenue – Total Costs – Surcharges. Income Tax = $\text{Max}(\text{Total Profit}, 0) \times 25\%$. Net Profit = Total Profit – Income Tax. Depreciation and amortization are non-cash expenses, so they are added back when computing operating net cash flow.

7.2 Evaluation Indicator Calculation Results

Based on the project investment cash flow statement, the three core evaluation indicators are calculated as follows:

NPV Calculation:

$$\begin{aligned}\text{NPV} &= -12.00 + 2.75/(1.10)^1 + 7.26/(1.10)^2 + 14.07/(1.10)^3 + 22.21/(1.10)^4 + 34.84/(1.10)^5 \\ &= -12.00 + 2.50 + 6.00 + 10.57 + 15.17 + 21.63 \\ &= 43.88 \text{ (RMB 10,000)} = \text{RMB 438,800}\end{aligned}$$

IRR Calculation:

IRR is the discount rate r such that $\text{NPV}(r) = 0$. Using Excel IRR function or iterative interpolation:
 $\text{NPV}(70\%) \approx +0.38$, $\text{NPV}(71\%) \approx -0.15 \Rightarrow \text{IRR} \approx 70.47\%$

Dynamic Investment Payback Period:

$$\begin{aligned}P_t' &= (T - 1) + |\text{Cumulative Discounted NCF at } (T-1)| / \text{Discounted NCF at Year } T \\ &= (3 - 1) + |-3.50| / 10.57 \\ &= 2 + 0.33 \\ &= 2.33 \text{ years}\end{aligned}$$

Indicator	Calculated Value	Judgment Criterion	Conclusion
Net Present Value NPV ($i_0 = 10\%$)	RMB 438,800	$\text{NPV} > 0$	Feasible — project creates excess returns
Internal Rate of Return IRR	70.47%	$\text{IRR} > i_0 (10\%)$	Feasible — significantly exceeds benchmark
Dynamic Investment Payback Period P_t'	2.33 years	$P_t' < 5 \text{ years}$	Feasible — capital recovered well within lifecycle
Cumulative Net Cash Flow (5 years)	RMB 691,300	Positive cumulative NCF	5-year total cash inflow covers initial investment

According to the above calculations, HeartCompass demonstrates strong profitability and investment recovery capability under the benchmark scenario. The revenue assumptions are oriented toward concrete digital companion persona scenarios: starting from personal subscriptions and small-group use cases, supplemented by customized commemorative archive services as ancillary revenue. The results are aligned with the project's current positioning as an MVP-stage AI companionship service.

7.2.1 Formal Judgment (per Step 3.3.2)

Following the judgment framework specified in the experiment instruction (Step 3.3.2): The above calculation results show that the project benefits meet all three acceptance criteria — $NPV = \text{RMB } 438,800 > 0$, $IRR = 70.47\% > i_0 = 10\%$, and $P_t' = 2.33 \text{ years} < 5\text{-year calculation period}$. Therefore, the project is judged FEASIBLE, and it is recommended to proceed to the next step: consider the financing plan and conduct post-financing analysis. Since this report adopts a pre-financing analysis framework and the benchmark scenario already demonstrates strong profitability, the project does not require design modification or abandonment. The sensitivity analysis below further confirms that even under adverse conditions, the project maintains positive NPV, reinforcing the judgment that HeartCompass is economically viable and ready for financing consideration.

7.3 Sensitivity Analysis

Given that user conversion rates and model invocation costs for digital companion persona products are inherently uncertain, this report selects revenue and operating cost as the two key variables for sensitivity analysis. Four scenarios are constructed:

Scenario	Assumptions	NPV (RMB 10,000)	IRR	P_t' (years)	Conclusion
Benchmark	Revenue and cost as per baseline	43.88	70.47%	2.33	Economically feasible
Conservative	Revenue -20%; Cost +10%	16.63	37.57%	3.47	Still feasible; NPV remains positive
Optimistic	Revenue +20%; Cost -5%	68.11	94.00%	1.88	Strong returns; rapid capital recovery
Stress Test	Revenue -30%; Cost +15%	2.75	15.29%	4.65	Near critical threshold; NPV still > 0

Sensitivity analysis insights: The most sensitive factors affecting project viability are paid conversion rate and model API invocation cost. If users only trial the product briefly without forming a long-term companionship relationship, subscription revenue will decline significantly. If users engage in high-frequency conversations without summary compression and invocation quota constraints, operating costs will escalate rapidly. Therefore, the immediate next-stage priorities should be validating retention rate, companionship effectiveness, and per-user model cost through seed-user testing.

Notably, even under the stress scenario (revenue -30% , cost $+15\%$), the NPV remains positive at RMB 27,500, and the IRR of 15.29% still exceeds the 10% benchmark. This indicates that HeartCompass possesses a reasonable margin of safety in its financial projections, largely attributable to its modest initial investment requirement and the high operating leverage characteristic of AI-native SaaS products.

VIII. Economic Feasibility and Risk Evaluation

8.1 Feasibility Conclusion

Based on the financial calculation results and the technical foundation established by the course project, this report concludes that HeartCompass possesses economic feasibility for continued advancement beyond the MVP stage. The main supporting reasons are:

1. Focused project positioning: HeartCompass is built around the "digital companion persona" rather than a generalized AI tool, making the target demand more concrete and addressable.
2. Modest initial investment: With a total investment of RMB 120,000, the project's capital requirement is manageable for seed-stage validation. Existing code, deployment workflows, bot entry points, and long-term memory prototypes are already delivered.
3. Revenue model aligned with genuine needs: Personal companionship, deep companionship, family/group sharing, and customized commemorative archives correspond to distinct paid tiers with differentiated value propositions.
4. Financial indicators meet acceptance criteria: Under the benchmark scenario, $NPV > 0$, $IRR (70.47\%) \gg \text{benchmark discount rate } (10\%)$, and $\text{dynamic payback period } (2.33 \text{ years}) < \text{project lifecycle } (5 \text{ years})$.

5. Adequate safety margin: Even under the conservative and stress scenarios, NPV remains positive, indicating that the project can withstand moderate adverse deviations.

8.2 Main Risks and Mitigation Measures

Risk Category	Specific Manifestation	Mitigation Strategy
Demand Risk	Users may find the digital companion persona novel initially but not form sustained usage habits	Validate real companionship needs through seed-user testing; iterate based on retention metrics
Cost Risk	High model API invocation frequency may compress margins	Implement invocation quotas, summary compression, prompt caching, and tiered model routing
Experience Risk	Unstable persona profiles or discontinuous responses degrade user trust	Optimize memory update algorithms, profile synchronization, and conversation quality monitoring
Privacy & Compliance Risk	System handles personal memories, chat records, and persona profiles	Strengthen authorization flows, data encryption, deletion mechanisms, and privacy documentation
Operations Risk	Long-term operation requires sustained server and database maintenance	Provide Dockerized one-click deployment, automated health checks, monitoring, and logging

IX. Thinking Question: Evaluating Software Projects Without Cash Income

If a software project generates no direct cash income, its value cannot be assessed solely through operating revenue, profit, and investment payback period. Instead, a combination of cost-benefit analysis, cost-effectiveness analysis, and multi-criteria comprehensive evaluation methods should be employed:

1. Cost-Saving Method (成本节约法): Estimate the labor hours, equipment costs, travel expenses, paper materials, and repeated communication overhead saved after the project is deployed. The saved cost is treated as the project's economic benefit. For example, an

internal automation tool may not generate revenue but can eliminate manual processing costs.

2. Efficiency-Improvement Method (效率提升法): Measure reductions in business process cycle time, error rates, and response latency, as well as increases in processing throughput. Convert these efficiency gains into monetary equivalents using labor cost rates.

3. Social Benefit Evaluation (社会效益评估): For public-welfare, educational, healthcare, and government projects, evaluate non-cash benefits such as service coverage expansion, user satisfaction improvement, risk reduction, public resource savings, and social equity enhancement.

4. Multi-Criteria Decision-Making (多准则决策): Apply methods such as the Analytic Hierarchy Process (AHP), expert scoring, and TOPSIS to comprehensively score the project across dimensions including security, usability, maintainability, social value, and strategic alignment.

5. Replacement-Cost Method (重置成本法): Estimate the cost of purchasing equivalent third-party systems, hiring additional personnel, or outsourcing services if the software were not developed. This avoided cost represents the implicit benefit of the project.

For a digital companion persona project such as HeartCompass, even in the absence of short-term direct cash income, its comprehensive benefits can still be evaluated across multiple dimensions: the preservation value of personal and family memories, the number of users covered by companionship services, user retention and engagement metrics, reduction in repeated communication overhead, improvement in personal knowledge reuse efficiency, and the degree of privacy risk mitigation. These non-monetary indicators collectively form a holistic picture of the project's value proposition and can serve as the basis for cost-effectiveness comparisons against alternative approaches.

X. Experimental Learning Experience

Through this experiment, I gained a systematic understanding that the economic evaluation of a software project is far more than a simple development cost estimate. It requires integrating the technical solution, user needs, business model, operating cost structure, tax policy, depreciation and amortization schedules, and multi-year cash flow projections into a single coherent financial model. For an AI-driven digital companion persona project like HeartCompass, what ultimately determines economic feasibility is not whether the project can invoke a large language model, but whether users can form a sustained companionship relationship with the system, and whether the per-user model inference cost can be controlled within viable margins.

This experiment also deepened my understanding of the interconnections among financial statements. The investment estimation table determines the Year 0 cash outflow. The revenue and cost tables determine operating profit. The tax schedule affects net profit. Depreciation and amortization reduce accounting profit but are non-cash items that must be added back in the cash flow statement. Finally, NPV, IRR, and the dynamic payback period are derived through discounted cash flow analysis. Understanding these linkages is essential for constructing a logically consistent and computationally correct evaluation model.

To make the parameter mapping relationships among financial tables explicit, I summarize the key data flows as follows: (1) The Investment Estimation Table (Table 5.2) feeds its Total Project Investment (RMB 120,000) into the Cash Flow Statement as the Year 0 outflow. (2) The Fixed Asset Depreciation Table and Intangible Asset Amortization Table (Table 5.4) feed annual depreciation (RMB 3,000) and amortization (RMB 13,000) into the Total Cost Table (Table 6.3) and are subsequently added back in the Cash Flow Statement as non-cash items. (3) The Operating Revenue Table (Table 6.1) feeds into both the VAT & Surcharge Table (Table 6.2) — where $VAT = Revenue \times 6\%$ and $Surcharges = VAT \times 12\%$ — and directly into the Cash Flow Statement as cash inflow. (4) $Total Profit = Operating Revenue - Total Costs - Surcharges$; $Income Tax = Total Profit \times 25\%$; $Net Profit = Total Profit - Income Tax$. (5) $Operating Net Cash Flow = Net Profit + Depreciation \& Amortization + Working Capital Recovery$. (6) NPV is the

sum of discounted net cash flows at $i = 10\%$; IRR is the rate where $NPV = 0$; the Dynamic Payback Period is the point where cumulative discounted net cash flow turns positive. Understanding this chain — from investment → cost/revenue → tax → profit → cash flow → NPV/IRR/Payback — is the core skill of software project economical evaluation.

From the perspective of the HeartCompass project specifically, the economic evaluation exercise made me more clearly aware that if the course project is to be further developed, the focus should not be on demonstrating technical stack completeness alone. Rather, the project must return to concrete user needs: why users need a digital companion persona, what memories they are willing to entrust to the system, what companionship capabilities they are willing to pay for, and how to balance privacy protection with cost control. The sensitivity analysis further reinforces that retention rate and unit economics (per-user model cost) are the two most critical operational metrics to validate in the next stage.

In terms of methodology, I learned to apply the NESMA function point analysis framework for software size measurement, the WBS-driven backward tracking method for cost validation, and the discounted cash flow model for investment decision-making. The dual-track cross-validation between NESMA FP (54 FP) and WBS hours (298 h) with only 5.20% variance was particularly instructive — it demonstrated that rigorous software measurement can produce highly consistent results across independent estimation approaches, which is essential for credible economic evaluation in both academic and industrial contexts.

Finally, this experiment helped me appreciate the importance of national standards and industry parameters in software engineering economics. The benchmark discount rate (10%), tax rates (6% VAT, 25% CIT), and depreciation/amortization rules (5-year straight-line) are not arbitrary choices but are grounded in the "Economic Evaluation Methods and Parameters for Construction Projects (Third Edition)" and current PRC tax regulations. Aligning the evaluation model with these standards ensures that the analysis is not only internally consistent but also comparable and auditable by external stakeholders.

References

- [1] National Development and Reform Commission, Ministry of Construction. Economic Evaluation Methods and Parameters for Construction Projects (Third Edition) [M]. Beijing: China Planning Press, 2006: 13-19.
- [2] China Software Industry Association. 2024 China Software Industry Benchmark Data (CSBMK-202410). Beijing, 2024.
- [3] NESMA. NESMA Function Point Analysis Standards — Comprehensive Functional Size Measurement Guidelines. Netherlands Software Metrics Association, 2018.
- [4] Public information on PRC tax administration: Corporate Income Tax Law, Value-Added Tax regulations, and related software service tax policies.
- [5] HeartCompass Course Project Repository. <https://github.com/Jackey0903/HeartCompass>.
- [6] HeartCompass Project Documentation: README.md, BRIEF_INTRO.md, WBS_TRACKING.md, and Biweekly Reports #1-#5.